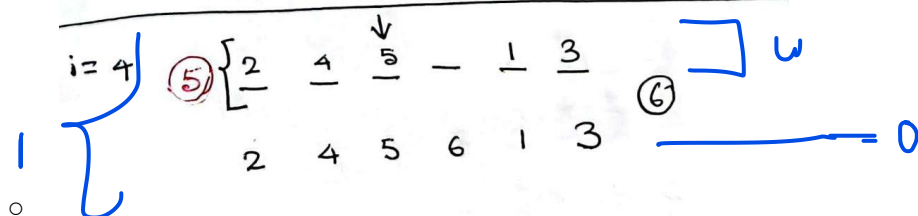
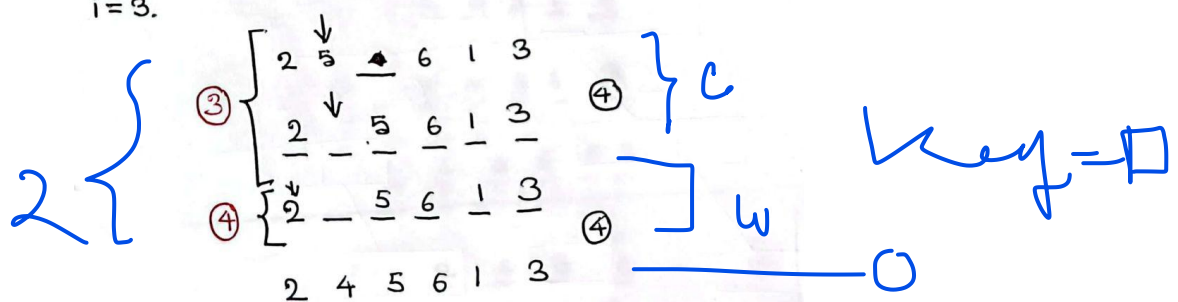
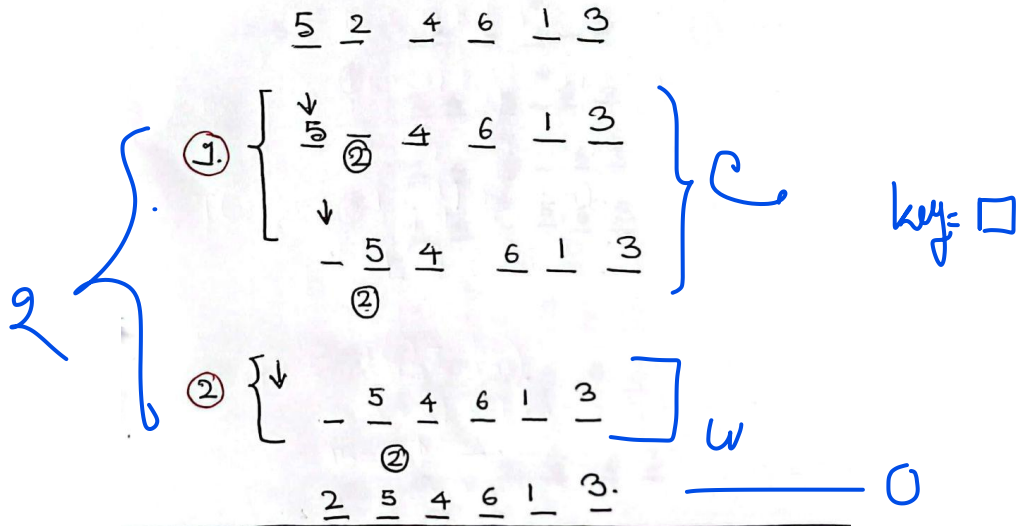


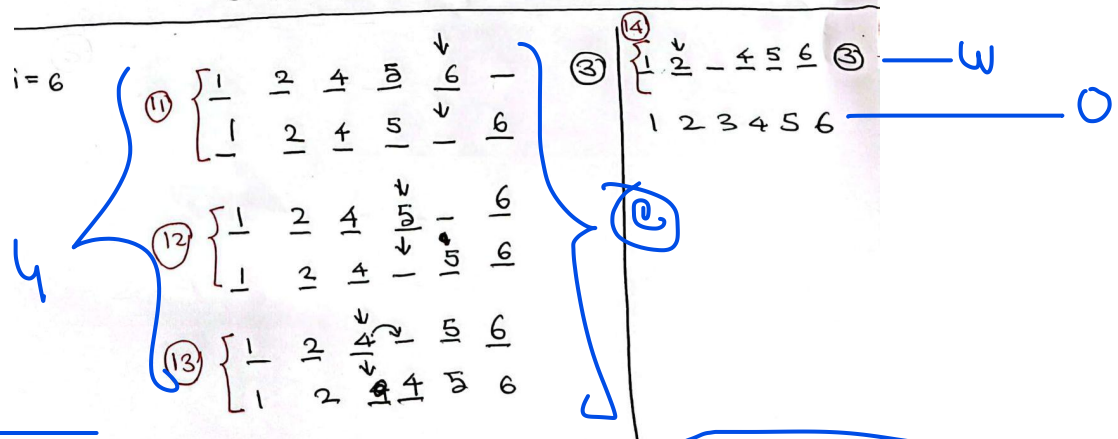
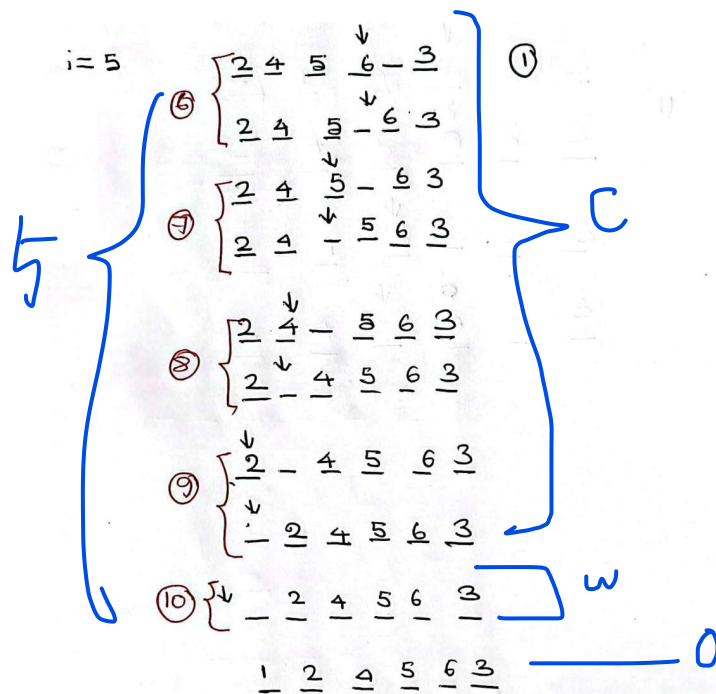
Insertion Sort

- For each iteration of the for loop, how many times the condition of the while loop condition will be checked if the following numbers are sorted in ascending order using insertion sort.

○ **Solution :**

$i = 2$





- Asymptotically what is the time required for insertion sort in best case/worst case scenario. Justify your answer.
 - Solution** : Done in class

Asymptotic Notation

- ✓ Show that the function $f(n) = 5n^3 + 10n^2 = O(n^3)$
 - Solution** : Done in class

Counting Sort

- The time Counting sort is denoted by an asymptotically tight bound but the time required by Insertion sort cannot be denoted in the same way. - Do you agree with the given statement? Give arguments in favor of your answer.
 - Solution** : Counting sort - same for all cases
- For the given sequence of numbers, what is the final value of the array C if counting sort is applied. Consider the value of k to be 5.
 - Solution**:

$$O(n+k)$$

Counting:

$$A: \begin{array}{ccccccc} 2 & 5 & 3 & 0 & 2 & 3 & 0 & 3 \\ \hline 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array}$$

$$C: \begin{array}{cccccc} 2 & 0 & 2 & 3 & 0 & 1 \\ \hline 0 & 1 & 2 & 3 & 4 & 5 \end{array}$$

Adding:

$$C: \begin{array}{cccccc} 2 & 2 & 4 & 7 & 7 & 8 \\ \hline 0 & 1 & 2 & 3 & 4 & 5 \end{array}$$

Finding B:

$$\begin{array}{ccccccc} A: & \begin{array}{c} 2 \\ \hline 1 \\ \uparrow \end{array} & \begin{array}{c} 5 \\ \hline 2 \\ \uparrow \end{array} & \begin{array}{c} 3 \\ \hline 3 \\ \uparrow \end{array} & \begin{array}{c} 0 \\ \hline 4 \\ \uparrow \end{array} & \begin{array}{c} 2 \\ \hline 5 \\ \uparrow \end{array} & \begin{array}{c} 3 \\ \hline 6 \\ \uparrow \end{array} & \begin{array}{c} 0 \\ \hline 7 \\ \uparrow \end{array} & \begin{array}{c} 3 \\ \hline 8 \\ \uparrow \end{array} \\ B: & \begin{array}{c} 0 \\ \hline 1 \end{array} & \begin{array}{c} 0 \\ \hline 2 \end{array} & \begin{array}{c} 2 \\ \hline 3 \end{array} & \begin{array}{c} 2 \\ \hline 4 \end{array} & \begin{array}{c} 3 \\ \hline 5 \end{array} & \begin{array}{c} 3 \\ \hline 6 \end{array} & \begin{array}{c} 3 \\ \hline 7 \end{array} & \begin{array}{c} 5 \\ \hline 8 \end{array} \end{array}$$

$$C: \begin{array}{cccccc} 2 & 2 & 4 & 7 & 7 & 8 \\ \hline 0 & 1 & 2 & 3 & 4 & 5 \end{array} \leftarrow \text{indices.}$$

Final C:

$$\begin{array}{cccccc} 0 & 2 & 2 & 4 & 7 & 7 \\ \hline 0 & 1 & 2 & 3 & 4 & 5 \end{array}$$

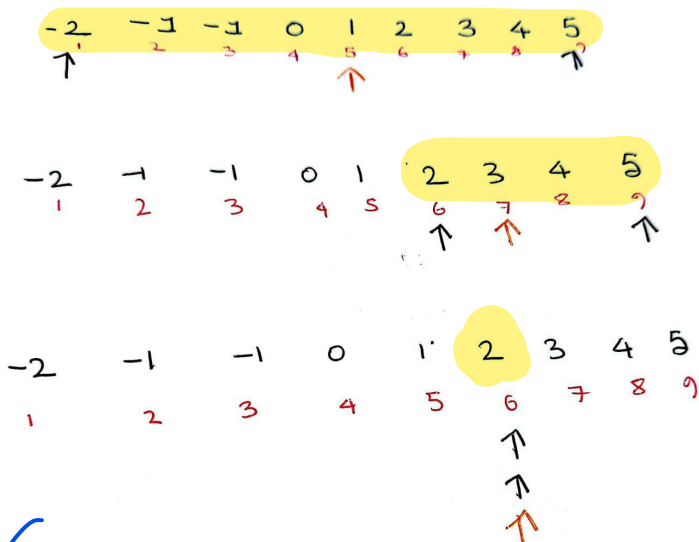
Array Mapping

- Address(arr[1][2]) in row major/col major order. Base index is [0][0]
- Solution** : Done in class

Binary Search

- Demonstrate how binary search works on the given array. Consider key = 2
- **Solution:**

key = 2



Merge Sort

- Demonstrate how merge sort will work on the given array.
 - Array : 7 2 9 4 3 8 6 1
 - **Solution:**

